

Machine Learning



Lesson Objectives

- Artificial Intelligence
 - AI Development
- Machine Learning
 - Applications of Machine Learnings

What is Artificial Intelligence?



"[The automation of] activities that we associate with human thinking, activities such as decision-making problem solving, learning ..." (Bellman, 1978)

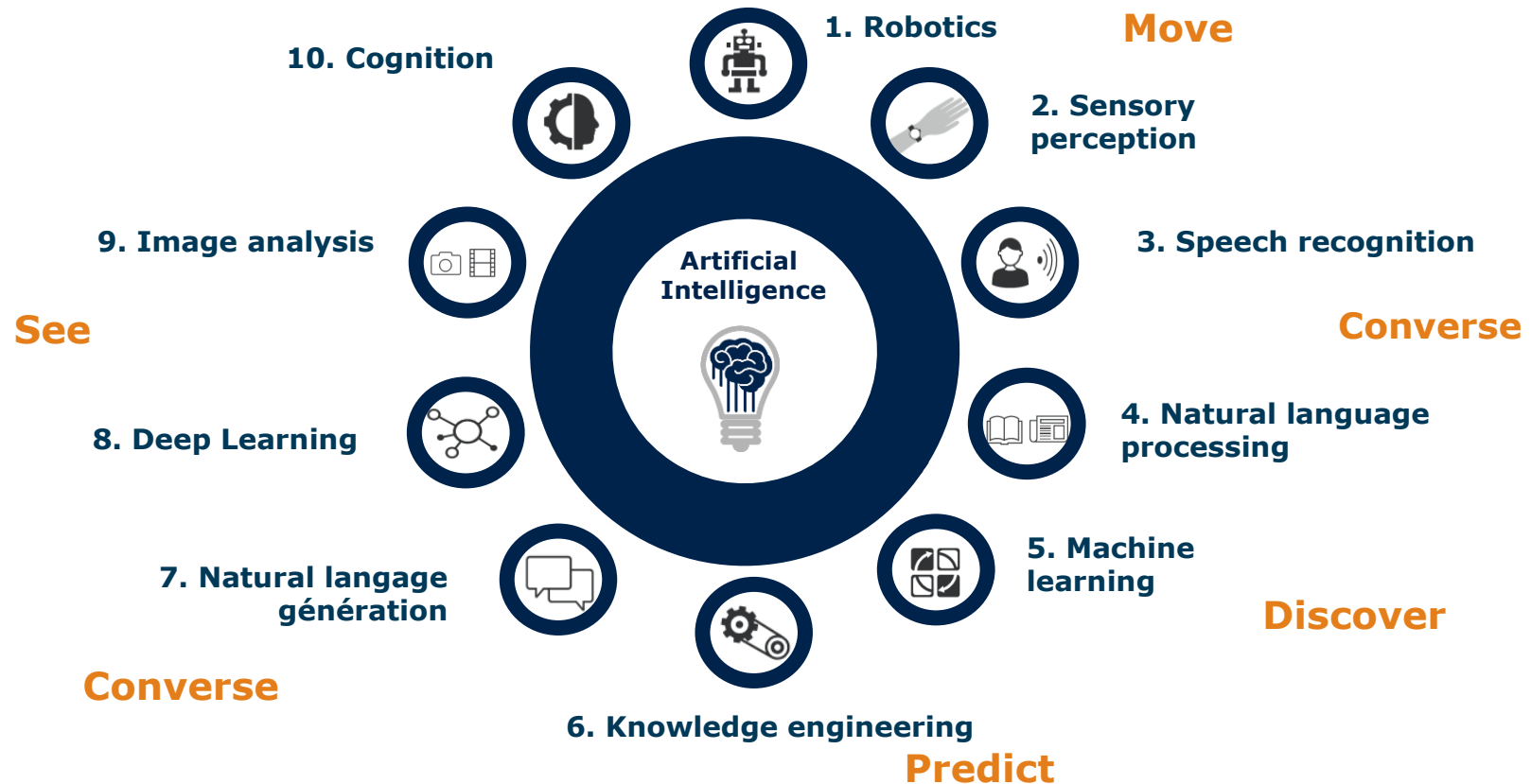
"A field of Study that seeks to explain and emulate intelligent behavior in terms of computational processes" (Schalkoff, 1990)

"The capability of a machine to immitate intelligent human behavior".
(merriam-wester)

Artificial intelligence, machine learning, deep learning and beyond

- Artificial intelligence (AI) brings with it a promise of genuine human-to-machine interaction.
- When machines become intelligent, they can understand requests, connect data points and draw conclusions.
- Now Machine can reason, observe and plan.
- Consider following examples;
 - **Leaving for a business trip tomorrow?** Your intelligent device will automatically offer weather reports and travel alerts for your destination city.
 - **Planning a large birthday celebration?** Your smart bot will help with invitations, make reservations and remind you to pick up the cake.
 - **Planning a direct marketing campaign?** Your AI assistant can instinctively segment your customers into groups for targeted messaging and increased response rates.

The era of Artificial Intelligence is now!





The history of AI and machine learning

- it didn't leap from single-player chess games straight into self-driving cars
- Artificial intelligence originally set out to make computers more useful and more capable of independent reasoning.
- Most historians trace the birth of AI to a Dartmouth research project in 1956 that explored topics like problem solving and symbolic methods.
- In the 1960s, the US Department of Defense took interest in this type of work and increased the focus on training computers to mimic human reasoning.
- Eg: the Defense Advanced Research Projects Agency (DARPA) completed street mapping projects in the 1970s. And DARPA produced intelligent personal assistants in 2003, long before Google, Amazon or Microsoft tackled similar projects.



AI Development

- Artificial Intelligence (1950s) – Giving intelligence to machine
- Machine Learning (1980s) – realizing artificial intelligence (speech recognition, image recognition, playing go, dialogue)
- Deep Learning (2006) – for machine learning for higher prediction accuracy
 - A powerful class of machine learning model
 - Modern reincarnation of artificial neural network (ANN)
 - Collection of simple, trainable mathematical functions



- **Machine learning** automates analytical model building. It uses methods from neural networks, statistics, operations research and physics to find hidden insights in data without being explicitly programmed where to look or what to conclude.

- **A neural network** is a kind of machine learning inspired by the workings of the human brain. It's a computing system made up of interconnected units (like neurons) that processes information by responding to external inputs, relaying information between each unit.



What is machine learning?

- A branch of **artificial intelligence**, concerned with the design and development of algorithms that allow computers to evolve behaviors based on empirical data.
- As intelligence requires knowledge, it is necessary for the computers to acquire knowledge.

“Machine Learning is programming computers to optimize a performance criterion using example data or past experience”

Introduction To Machine Learning by E. Alpaydin

It is subset of Artificial Intelligence

“Machine learning is a core, transformative way by which we’re rethinking everything we’re doing” — Sundar Pichai, Google CEO



- The success of machine learning system also depends on the algorithms.
- The algorithms control the search to find and build the knowledge structures.
- The learning algorithms should extract useful information from training examples.



- **Supervised learning**

- Prediction
- Classification (discrete labels), Regression (real values)

- **Unsupervised learning**

- Clustering
- Probability distribution estimation
- Finding association (in features)
- Dimension reduction

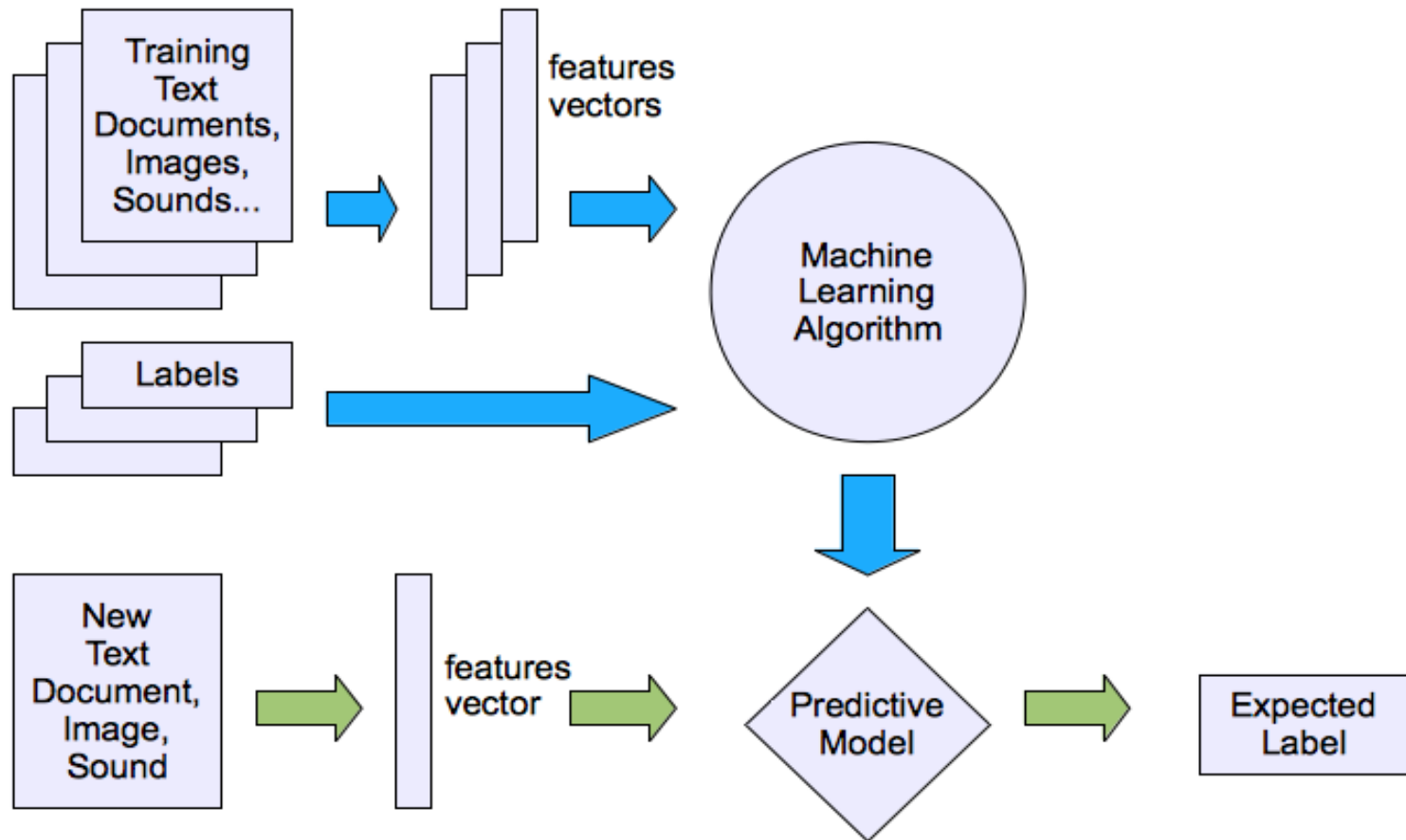
- **Semi-supervised learning Reinforcement learning**

- Decision making (robot, chess machine)

Machine learning structure



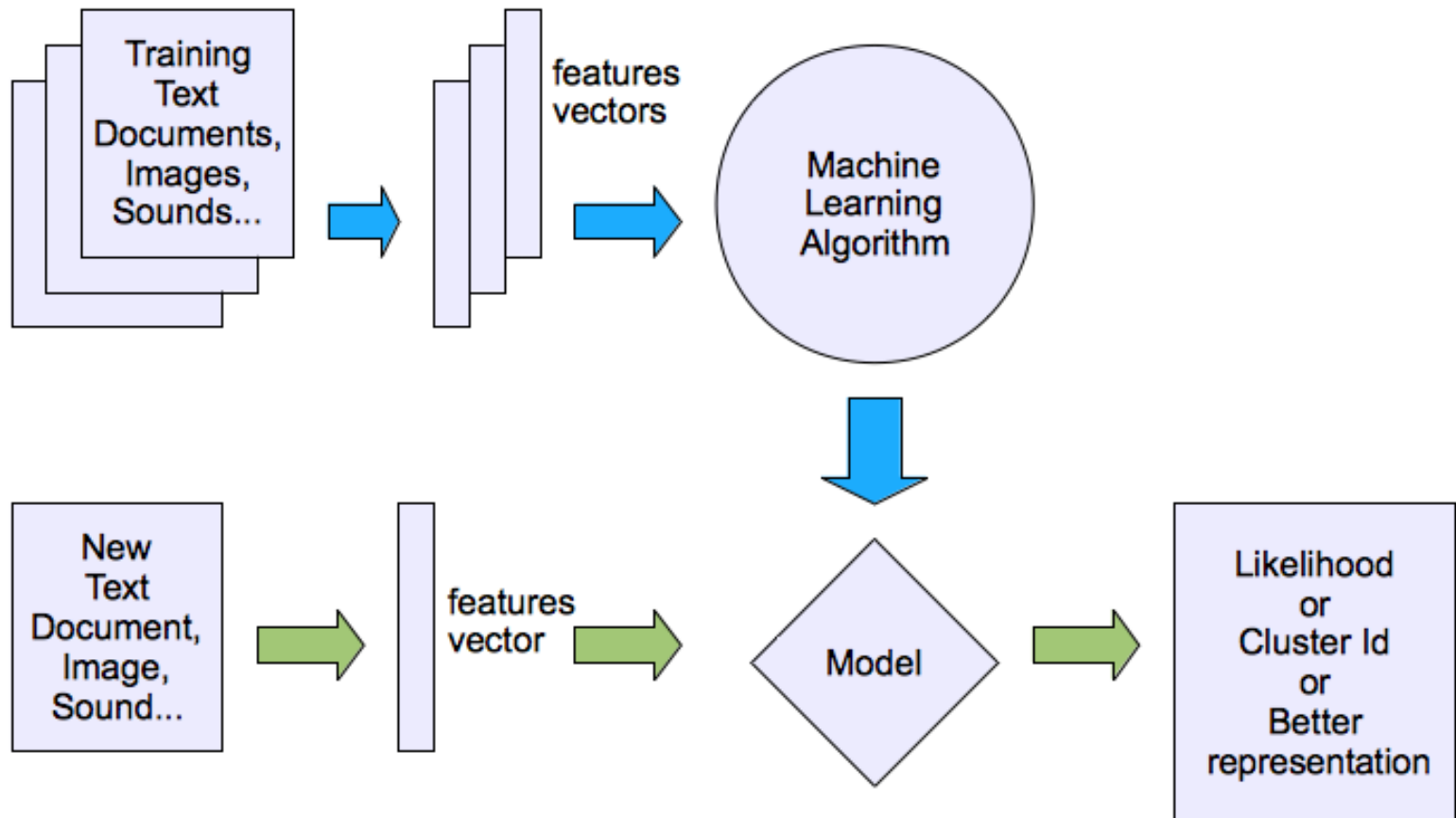
Supervised learning





Machine learning structure

Unsupervised learning



Machine Learning- Few Use Cases



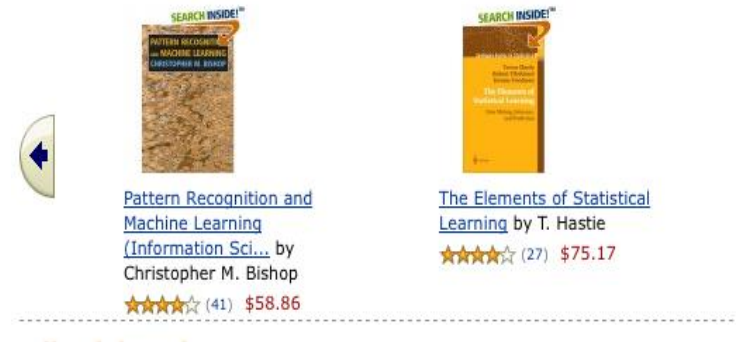
- Recommendation
- Clustering
- Classification
- Frequent pattern mining



Machine Learning Use Cases - Recommendations

- Extensive framework for collaborative filtering
- Recommenders
 - User based
 - Item based
- Online and Offline support
 - Offline can utilize Hadoop
- Many different Similarity measures (algorithms)
 - Cosine, LLR, Tanimoto, others

Customers Who Bought This Item Also Bought





- Document level
 - Group documents based on a notion of similarity
 - K-Means, Fuzzy K-Means, Dirichlet, Canopy, Mean-Shift
 - Distance Measures
 - Manhattan, Euclidean, other algorithms

[McCain the show horse: Way off track](#)

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[Russian Navy ship sent to combat pirates](#) ABC Online

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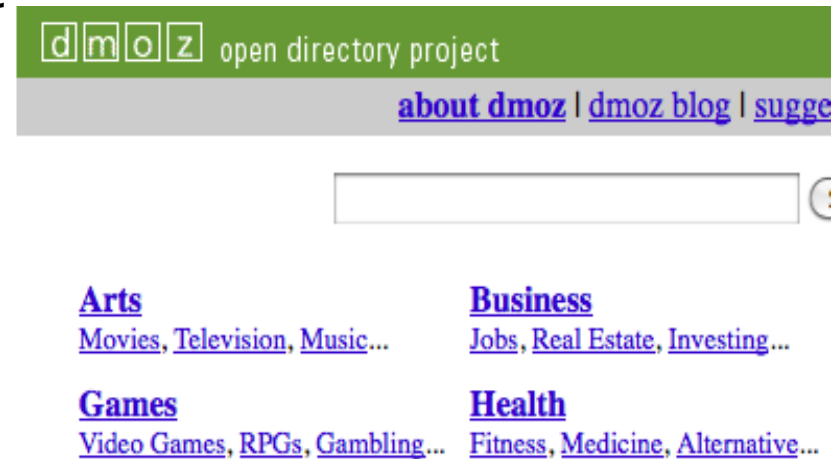


- Topic Modeling
 - Cluster words across documents to identify topics (trends, tag clouds)
 - Latent Dirichlet Allocation (LDA)



Categorization

- Place new items into predefined categories:
 - Sports, politics, entertainment
- Mahout has several implementations
 - Naïve Bayes
 - Complementary Naïve Bayes
 - Decision Forests
 - Support vector machines





Frequent Pattern Mining

- Identify frequently co-occurrent items
- Useful for:
 - Query Recommendations
 - Apple -> iPhone, Orange, OS X
 - Did you meant xxx?

<http://www.amazon.com>

"apple"

Related Searches: [apple iphone](#), [iphone](#), [ipod](#).

Select Results from All Departments

Patterns of interest



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[Pattern Recognition and Machine Learning \(Information Sci...](#) by Christopher M. Bishop
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Differences at nut shell



Artificial Intelligence

Machine Learning

Deep Learning

The subset of machine learning composed of algorithms that permit software to train itself to perform tasks, like speech and image recognition, by exposing multilayered neural networks to vast amounts of data.

A subset of AI that includes abstruse statistical techniques that enable machines to improve at tasks with experience. The category includes deep learning

Any technique that enables computers to mimic human intelligence, using logic, if-then rules, decision trees, and machine learning (including deep learning)



Difference between Machine Learning and Deep Learning

Sr.No	Parameters	Machine Learning	Deep Learning
1	Data Dependencies	Excellent performances on a small/medium dataset	Excellent performance on a big dataset
2	Hardware dependencies	Work on a low-end machine.	Requires powerful machine, preferably with GPU: DL performs a significant amount of matrix multiplication
3	Feature engineering	Need to understand the features that represent the data	No need to understand the best feature that represents the data
4	Execution time	From few minutes to hours	Up to weeks. Neural Network needs to compute a significant number of weights
5	Interpretability	Some algorithms are easy to interpret (logistic, decision tree), some are almost impossible (SVM, XGBoost)	Difficult to impossible

When to use ML or DL?



Sr.No	Parameters	Machine learning	Deep learning
1	Training dataset	Small	Large
2	Choose features	Yes	No
3	Number of algorithms	Many	Few
4	Training time	Short	Long

Source: <https://www.guru99.com/machine-learning-vs-deep-learning.html>



References

- <https://medium.com/iotforall/the-difference-between-artificial-intelligence-machine-learning-and-deep-learning-3aa67bff5991>
- <https://www.geospatialworld.net/blogs/difference-between-ai%EF%BB%BF-machine-learning-and-deep-learning/>
- <https://www.guru99.com/machine-learning-vs-deep-learning.html>



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